

Physical Chemistry Practical 3: Thermochemistry & Calorimetry

Enthalpy of Displacement with Cooling-Curve Extrapolation & Hess's Law Investigations

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Physical Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Determination of the Enthalpy Change of Displacement of Copper by Zinc

Cambridge Syllabus Reference: 9701/31/O/N/23/Q2 · [22 Marks]

Chemicals and Apparatus Provided:

- FA 1 is 1.00 mol dm^{-3} copper(II) sulfate, CuSO_4 .
- FA 2 is zinc powder, Zn .
- Polystyrene cup supported in a 250 cm^3 glass beaker.
- Thermometer (-10°C to $+110^\circ\text{C}$, reading to 0.5°C).

Experimental Procedure:

1. Using a measuring cylinder, transfer exactly 50.0 cm^3 of 1.00 mol dm^{-3} CuSO_4 (FA 1) into the polystyrene cup.
2. Weigh 3.50 g of zinc powder (FA 2) in a weighing bottle.
3. Place the thermometer in the cup, stir gently, and record the temperature every 30 seconds for 2.5 minutes.
4. At exactly $t = 3.0$ minutes, add the zinc powder all at once. DO NOT record the temperature at 3.0 minutes. Stir continuously with the thermometer.
5. Continue recording the temperature every 30 seconds from $t = 3.5$ minutes to $t = 10.0$ minutes.

(a) Record your temperature readings at 30-second intervals in the table below. [4]

Time / min	Temp / $^\circ\text{C}$	Time / min	Temp / $^\circ\text{C}$	Time / min	Temp / $^\circ\text{C}$
0.0		3.5		7.0	
0.5		4.0		7.5	
1.0		4.5		8.0	
1.5		5.0		8.5	
2.0		5.5		9.0	
2.5		6.0		9.5	
3.0 (mix)	X	6.5		10.0	

(b) On the grid provided, plot a graph of temperature (vertical axis) against time (horizontal axis). Draw two lines of best fit: one through the initial points (0 to 2.5 min) and one through the cooling points (4.0 to 10.0 min). Extrapolate both lines to $t = 3.0$ minutes to determine the theoretical maximum temperature change, T . [6]

Thermochemistry: Temperature vs Time Cooling Curve with Extrapolation (Cambridge 9701)

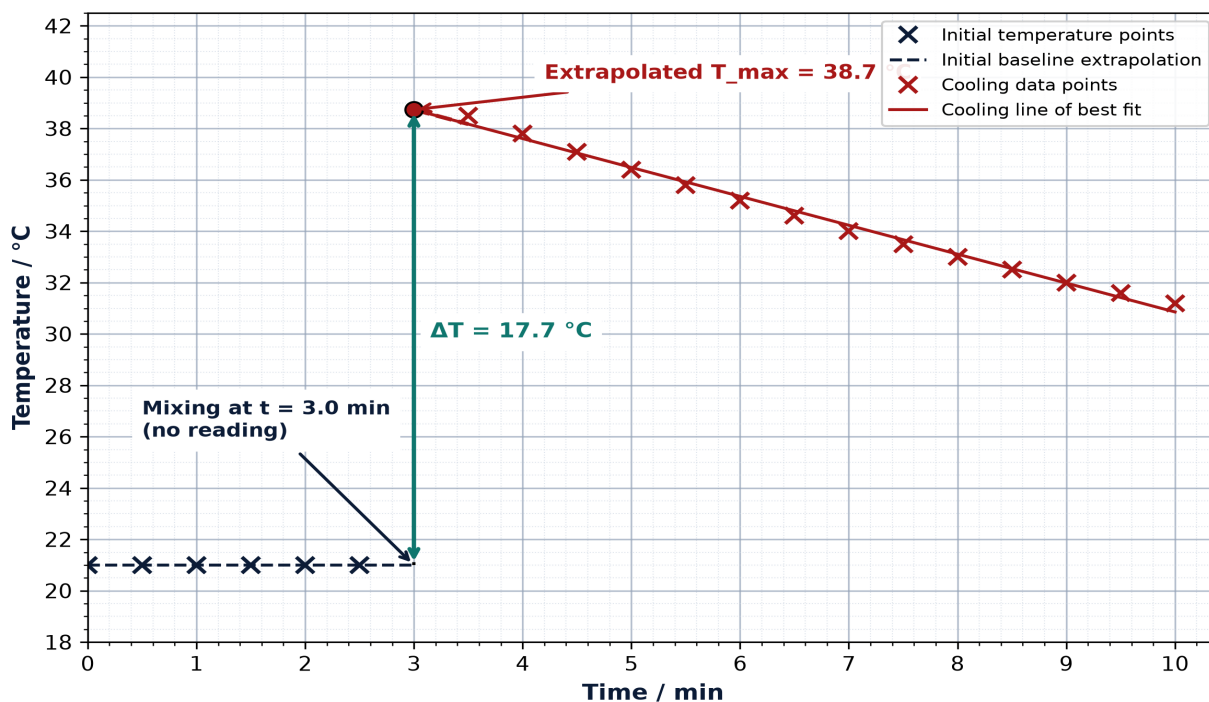


Fig 1.1: Temperature vs Time cooling curve with linear extrapolation to mixing time ($t = 3.0$ min).

(c) State the value of T obtained from your extrapolated graph. [1]

(d) Calculate the heat energy, q , evolved during the reaction in the polystyrene cup. [Specific heat capacity of solution = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$; Density of solution = 1.00 g cm^{-3}] [2]

(e) Calculate the amount, in moles, of CuSO_4 used and the amount, in moles, of Zn used. State which reagent is in excess. [Ar: Zn = 65.4] [3]

(f) Calculate the molar enthalpy change of displacement, H , in kJ mol^{-1} . Include the correct sign and express your answer to three significant figures. [2]

(g) The thermometer used has an uncertainty of ± 0.5 °C per reading. Calculate the percentage apparatus uncertainty in your value of T . [2]

(h) Explain why extrapolating the cooling curve to the time of mixing ($t = 3.0$ min) gives a more accurate value for T than simply recording the highest temperature reached on the thermometer. [2]

Question 2: Calorimetric Enthalpy of Neutralization: Strong Acid vs Strong Base**Cambridge Syllabus Reference: 9701/33/O/N/23/Q2 · [18 Marks]**

Chemicals and Apparatus Provided:

- FA 3 is 2.00 mol dm^{-3} hydrochloric acid, HCl.
- FA 4 is 2.00 mol dm^{-3} sodium hydroxide, NaOH.
- Two polystyrene cups with lids.
- Measuring cylinder (50 cm^3) and thermometer.

Experimental Procedure:

1. Measure 25.0 cm^3 of 2.00 mol dm^{-3} HCl (FA 3) and place into cup 1. Record its temperature T_1 .
2. Measure 25.0 cm^3 of 2.00 mol dm^{-3} NaOH (FA 4) and place into cup 2. Record its temperature T_2 .
3. Calculate the average initial temperature $T_{\text{initial}} = (T_1 + T_2) / 2$.
4. Pour the NaOH quickly into the HCl cup, place the lid, stir with the thermometer, and record the highest temperature reached, T_{max} .

(a) Record your temperature values T_1 , T_2 , T_{initial} , and T_{max} , and calculate the temperature rise T . [3]

.....

.....

.....

.....

(b) Calculate the heat energy evolved, q , in Joules. [Total volume = 50.0 cm^3 ; density = 1.00 g cm^{-3} ; $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$] [2]

.....

.....

.....

(c) Calculate the amount, in moles, of water formed during this neutralization reaction: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$. [2]

.....

.....

.....

(d) Calculate the standard enthalpy change of neutralization, H_{neut} , in kJ mol^{-1} . Include the sign and quote to 3 significant figures. [2]

.....

.....

.....

(e) The literature value for H_{neut} is $-57.1 \text{ kJ mol}^{-1}$. Calculate the percentage difference between your experimental result and the literature value. [2]

.....

.....

.....

(f) State two modifications to the apparatus that would reduce heat loss and improve the accuracy of the result. [2]

.....

.....

.....

(g) Predict how the value of H_{neut} would differ if 2.00 mol dm^{-3} ethanoic acid (a weak acid) were used instead of hydrochloric acid. Explain your reasoning. [3]

.....

.....

.....

.....
.....

(h) Suggest why using a glass beaker instead of a polystyrene cup produces a significantly larger error in calorimetric experiments. [2]

.....
.....
.....

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Physical Chemistry Practical 3: Thermochemistry & Calorimetry

Question 1: Determination of the Enthalpy Change of Displacement of Copper by Zinc [22 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Table formatted correctly, time in minutes, temperature recorded to nearest 0.5 °C, all 14 readings recorded [4].	[4]
(b)	Axes correctly labeled with quantity and unit (Temperature / °C, Time / min) [1] Scale chosen so plotted points occupy $\geq 50\%$ of each axis [1] All points plotted accurately within half a small square [1] Line of best fit drawn for pre-mixing points (horizontal baseline) [1] Line of best fit drawn for cooling points (linear descent) [1] Correct extrapolation of both lines to $t = 3.0$ min and accurate reading of T [1]	[6]
(c)	T recorded to nearest 0.5 °C (e.g. T = 18.5 °C).	[1]
(d)	Mass of solution $m = 50.0$ g. $q = m \times c \times \Delta T = 50.0 \times 4.18 \times 18.5 = 3,866.5$ J = 3.867 kJ. [2]	[2]
(e)	Moles of $\text{CuSO}_4 = 1.00 \text{ mol dm}^{-3} \times (50.0 / 1000) = 0.0500$ mol [1] Moles of Zn = $3.50 \text{ g} / 65.4 \text{ g/mol} = 0.0535$ mol [1] $0.0535 > 0.0500$, therefore Zn is in excess and CuSO_4 is the limiting reagent [1]	[3]
(f)	$H = -(q / \text{moles limiting}) = -(3.867 \text{ kJ} / 0.0500 \text{ mol}) = -77.3 \text{ kJ mol}^{-1}$. (Must include negative sign and correct 3 s.f.). [2]	[2]
(g)	Two temperature readings taken, total uncertainty = $2 \times 0.5 = 1.0$ °C. % uncertainty = $(1.0 / T) \times 100\% = (1.0 / 18.5) \times 100\% = 5.41\%$. [2]	[2]
(h)	The reaction is not instantaneous and heat is continuously lost to the surroundings from the moment reaction begins. The maximum temperature reached on the thermometer is lower than the true theoretical temperature because cooling has already occurred. Extrapolating the cooling curve backwards compensates for heat lost during the initial reaction phase. [2]	[2]

Question 2: Calorimetric Enthalpy of Neutralization: Strong Acid vs Strong Base [18 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	T1 and T2 recorded to 0.5 °C [1]; T _{max} recorded [1]; T correctly calculated (e.g. T _{initial} = 21.0 °C, T _{max} = 34.5 °C, T = 13.5 °C) [1].	[3]
(b)	$q = m \times c \times \Delta T = 50.0 \times 4.18 \times 13.5 = 2,821.5$ J = 2.822 kJ. [2]	[2]
(c)	Moles of HCl = $2.00 \times (25.0 / 1000) = 0.0500$ mol. Moles of NaOH = 0.0500 mol. Moles of H ₂ O formed = 0.0500 mol. [2]	[2]
(d)	$H_{\text{neut}} = -(2.822 \text{ kJ} / 0.0500 \text{ mol}) = -56.4 \text{ kJ mol}^{-1}$. (Exothermic sign mandatory, 3 s.f.). [2]	[2]
(e)	% difference = $(56.4 - 57.1 / 57.1) \times 100\% = (0.7 / 57.1) \times 100\% = 1.23\%$. [2]	[2]
(f)	1. Nest two polystyrene cups together (double-cupping) or place mineral wool insulation between cups. [1] 2. Use a tight-fitting lid with a small hole for the thermometer to prevent convective heat loss and vapor escape. [1]	[2]
(g)	The value of H_{neut} would be less exothermic (less negative, $\sim -54 \text{ kJ mol}^{-1}$). Ethanoic acid is only partially dissociated in aqueous solution. Energy must be absorbed from the surroundings to break covalent O-H bonds and fully dissociate the weak acid molecules into H ⁺ and ethanoate ions, reducing the net heat released during neutralization. [3]	[3]
(h)	Glass has a much higher thermal conductivity than expanded polystyrene and a larger heat capacity, absorbing and dissipating significant heat to the air, causing a substantially lower recorded T. [2]	[2]